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STUDENT PERFORMANCE TRACKER :

- ✓ It manages students, their grades, and generates analytics — all from the command line.

The 7 sections and what each does

Section 1 — Models Two core classes: GradeRecord stores a single grade entry (subject, score, term, date, notes) and converts scores to letter grades (A–F). Student holds a list of grade records and supports iteration, filtering by subject, rolling averages, and best/worst record lookups.

Section 2 — Exceptions A 10-class custom exception hierarchy. Examples: ScoreOutOfRangeError (score not in 0–100), StudentNotFoundError, DuplicateStudentError, CorruptDataError. This makes error messages precise and meaningful.

Section 3 — Pipeline Uses functional programming (higher-order functions, reduce) and a ScoreQueue built on Python's deque to compute a sliding moving average of the last 5 scores per student.

Section 4 — Storage Data is saved/loaded as JSON (with atomic writes and automatic backups), exportable to CSV, and printable as a plain-text report. An audit.log records every grade entry ever made.

Section 5 — Design Patterns (3 GoF)

- **Observer** — notifies listeners whenever a grade is added (e.g. for logging)
- **Strategy** — swappable grading scales (Standard, Strict, Lenient)

- **Decorator** — wraps a Student with extra behaviour like GPA tracking or read-only enforcement (archived students)

Section 6 — Registry A central service (REGISTRY) that manages all students — adding, removing, archiving, looking up, computing class averages, and importing CSV data.

Section 7 — UI / Main Menu A colour-coded terminal interface with 10 menu options:

Option

- 1 Add Student
- 2 Add Grade Record
- 3 View Student Report
- 4 Class Overview (ranked leaderboard)
- 5 Subject Analysis (avg, median, pass rate)
- 6 Change Grading Scale
- 7 Archive / Unarchive Student
- 8 Import from CSV
- 9 Export to CSV or text report
- 10 Delete Student